CMAI 480

Computer Vision

Dr. Khalil Ahmad Yousef
Computer Science and Engineering Department
Qatar University

Image Alignment and Stitching

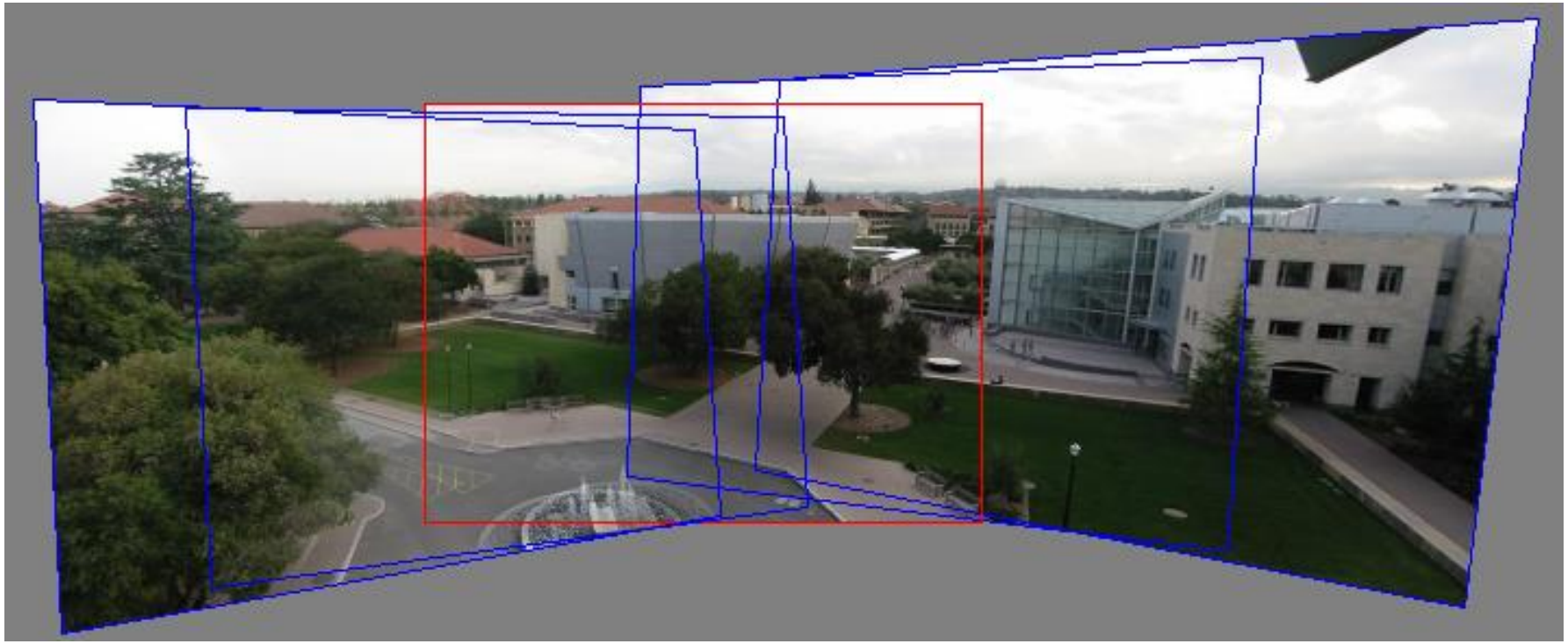
Chapter 8

Panoramas: Main Idea

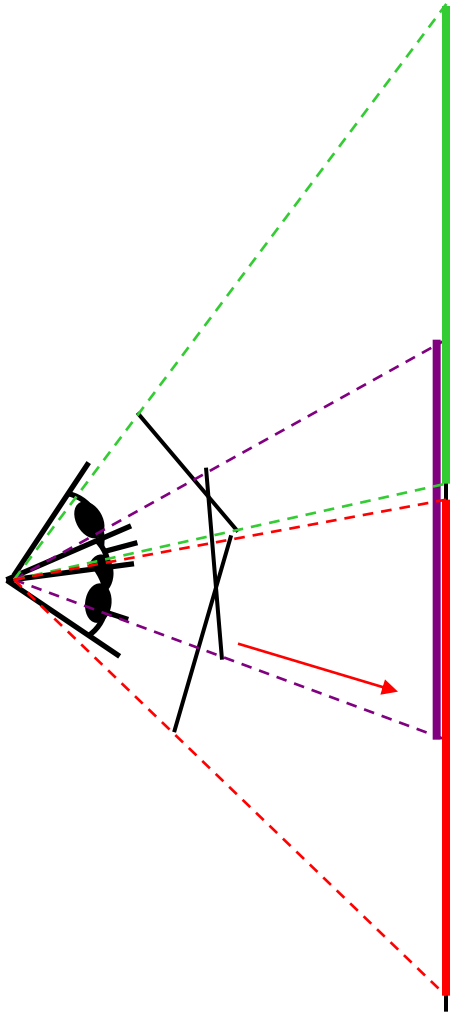
- Given two images:
  - Step 1: Detect features
  - Step 2: Match features
  - Step 3: Compute a homography using RANSAC
  - Step 4: Combine the images together (somehow)

- What if we have more than two images?

Can we use homographies to create a 360 panorama?

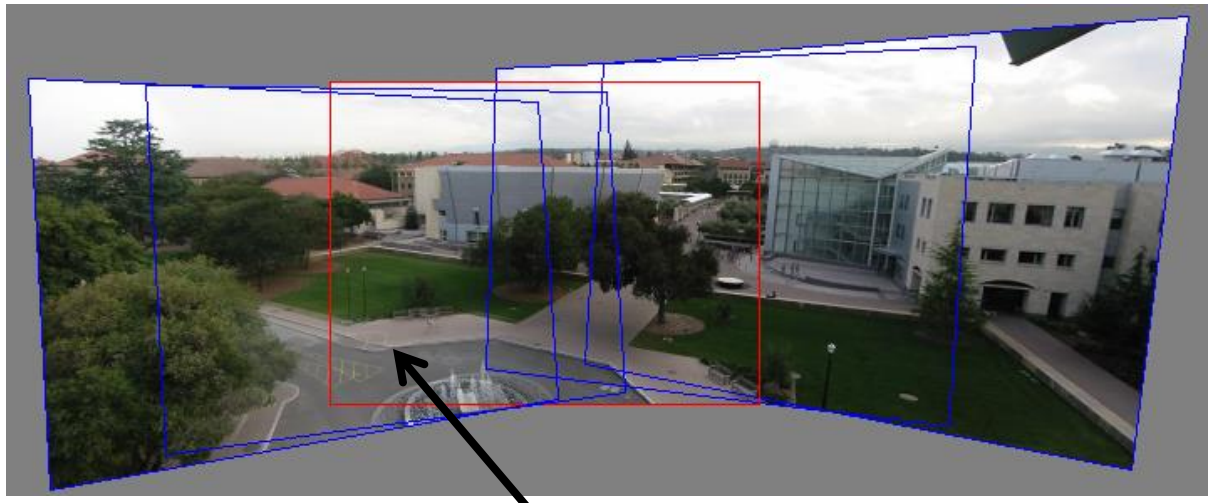


Idea: project images onto a common plane



mosaic projection plane

"Mosaic"



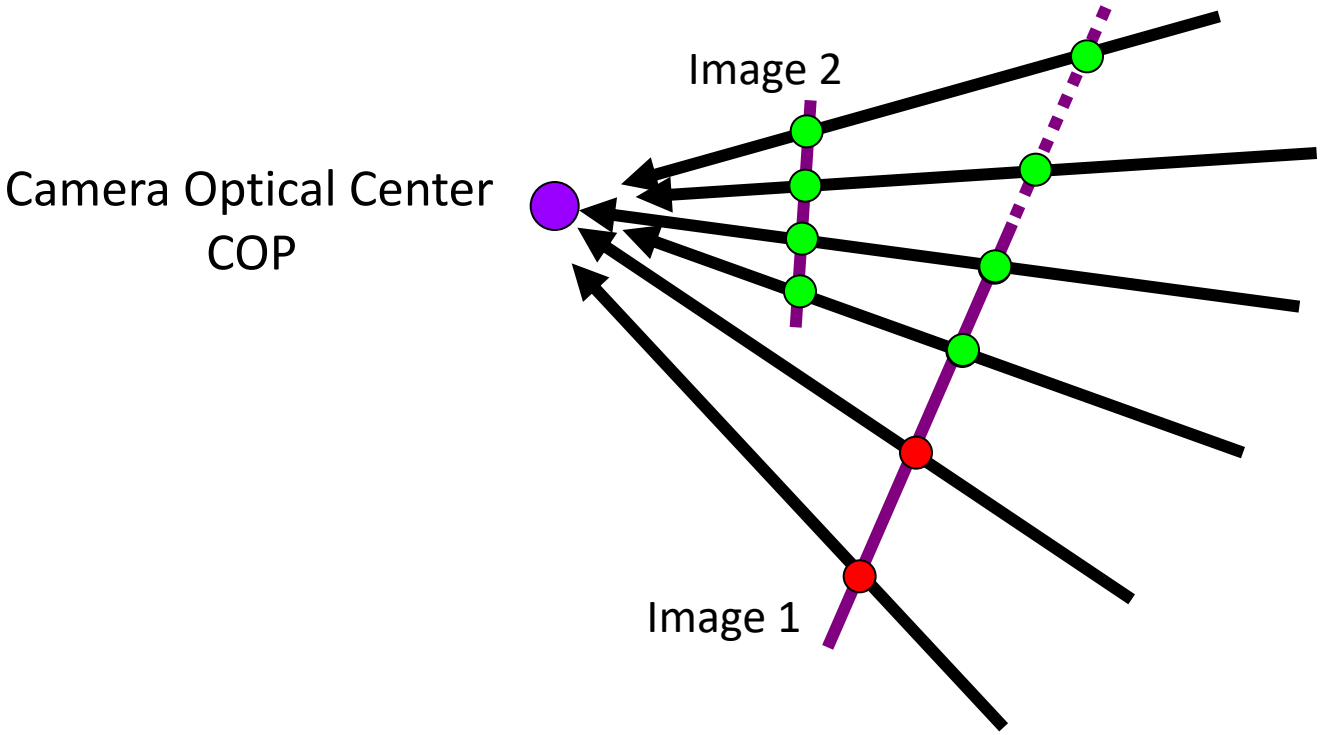
each image is warped
with a homography $\mathbf{H}$

Can't create a 360 panorama this way… we'll fix this shortly

Creating a panorama

- Basic Procedure
  - Take a sequence of images from the same position
    - Rotate the camera about its optical center
  - Compute transformation between second image and first
  - Transform the second image to overlap with the first
  - Blend the two together to create a mosaic
  - If there are more images, repeat

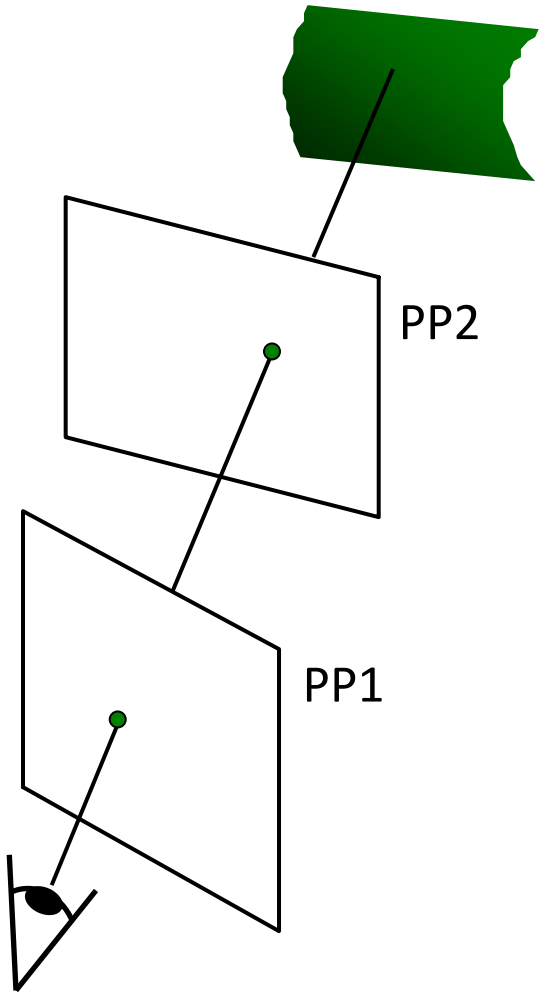
Geometric interpretation of mosaics



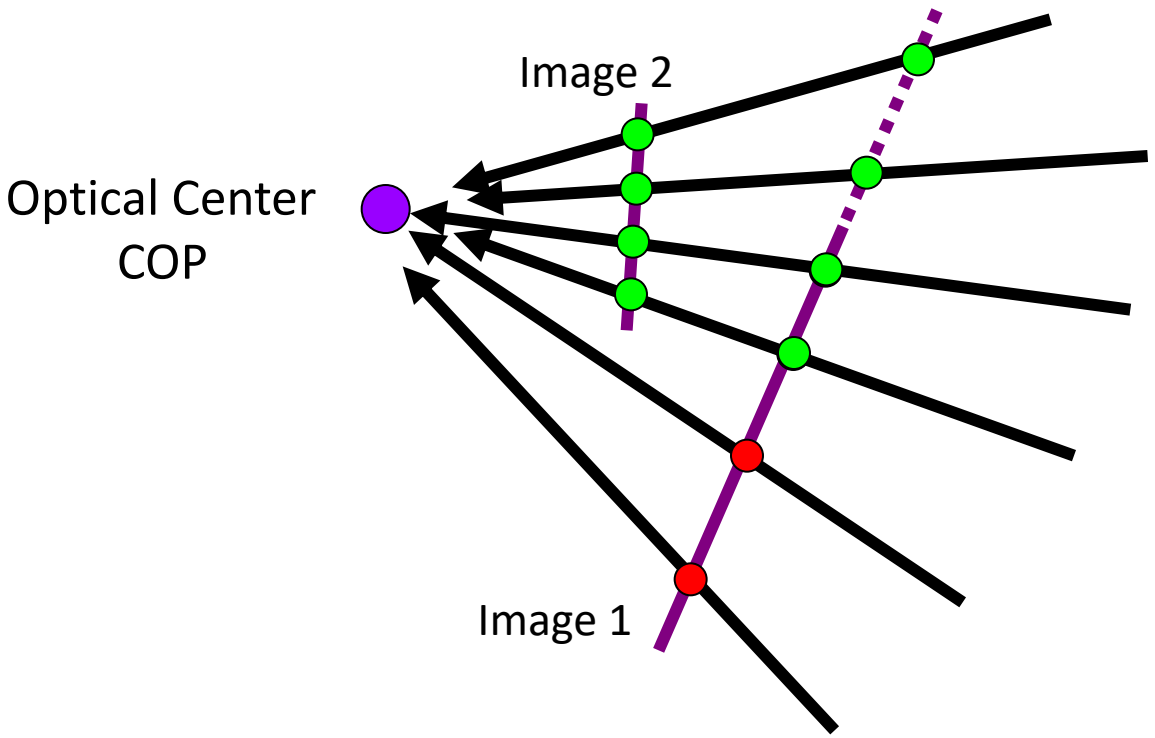
- If we capture all 360° of rays, we can create a 360° panorama
- The basic operation is *projecting* an image from one plane to another
- The projective transformation is scene-INDEPENDENT
  - This depends on all the images having the same optical center

Image reprojection

- Basic question
  - How to relate two images from the same camera center?
    - how to map a pixel from PP1 to PP2
- Answer
  - Cast a ray through each pixel in PP1 (Backprojection)
  - Draw the pixel where that ray intersects PP2



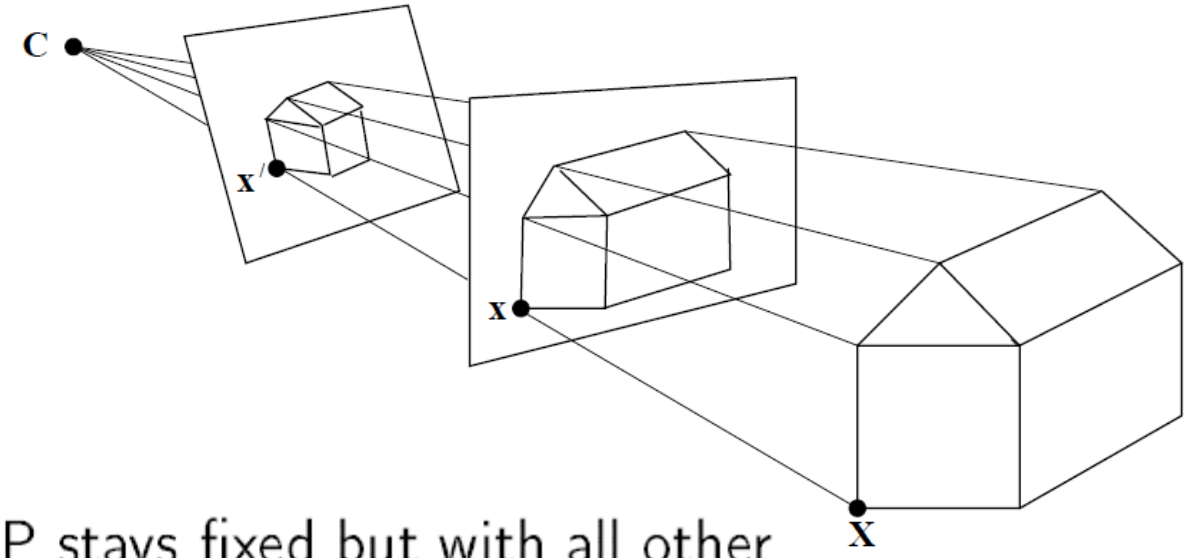
What is the transformation?



How do we map points in image 2 into image 1?

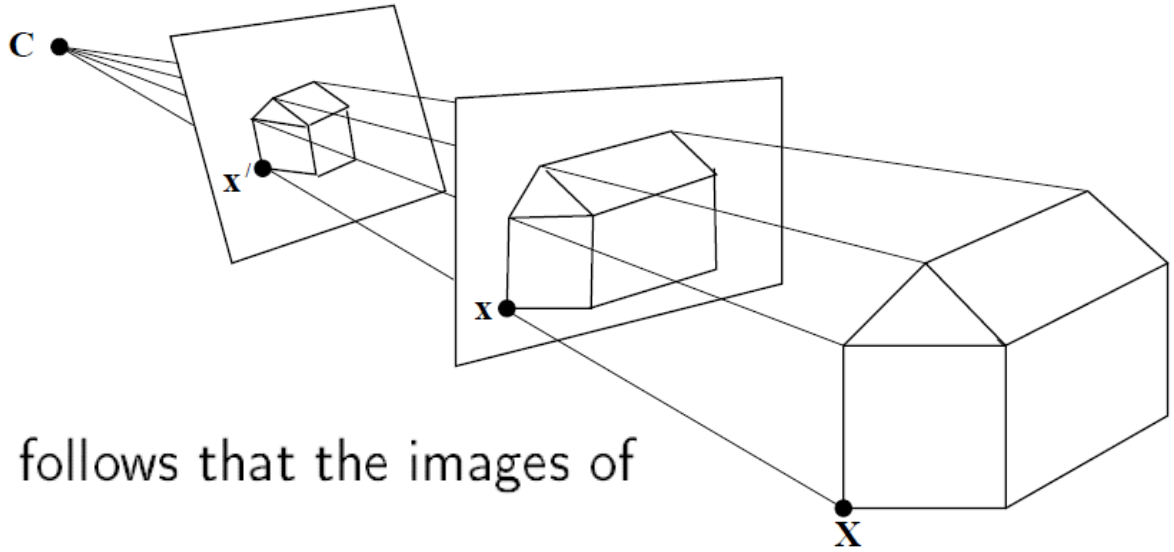
	image 1	image 2
intrinsics	$\boldsymbol{K_1 = K}$	$\boldsymbol{K_2 = K}$
extrinsics (rotation only)	$\boldsymbol{R_1 = I_{3\times 3}}$	$\boldsymbol{R_2 = R}$

Mosaicking



- **Intuition:** Imaging with a camera, where CoP stays fixed but with all other parameters are all allowed to vary or change
  - Assume pinhole camera model: $P = KR\left[I|-\tilde{C}\right]$
  - Let's assume the same view in the world space imaged by a camera with different orientation.
  - Both of the cameras should captures the same world point $\vec{X}$
    - First camera: $\vec{x} = P\vec{X} = KR\left[I|-\tilde{C}\right]\vec{X}$
    - Second camera: $\vec{x}' = P'\vec{X}' = K'R'\left[I|-\tilde{C}\right]\vec{X}'$
    - Note that $\vec{X} = \vec{X}'$ and $K = K'$
    - If we insist that K is not necessary equal to K', and since the cameras have a common center, then: there is a simple relation between P and P' namely $P' = (K'R')(KR)^{-1}P$ (Fact)

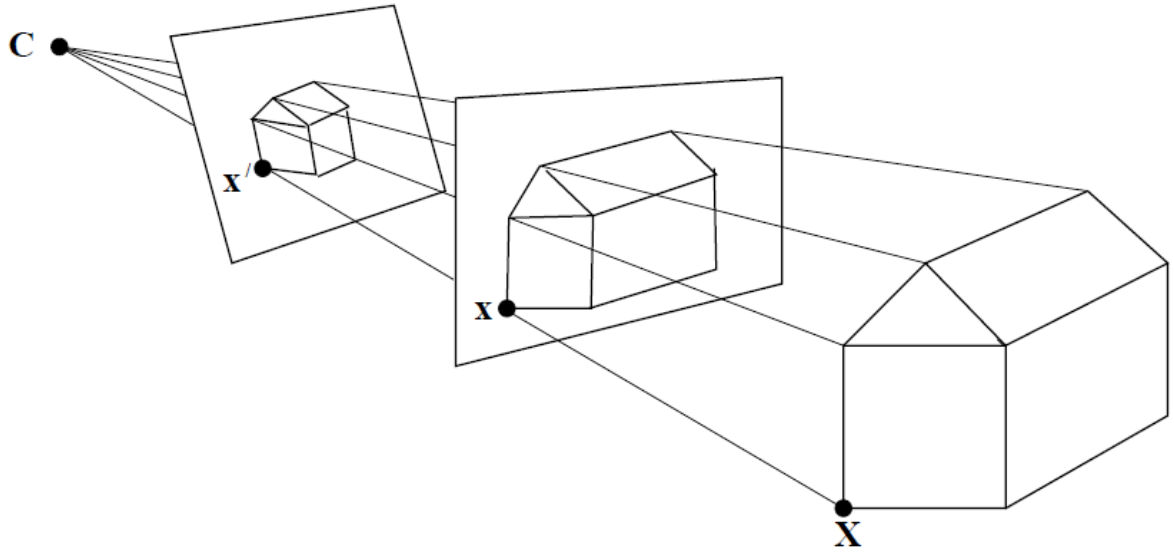
Mosaicking



- Given the fact that $P' = (K'R')(KR)^{-1}P$, it then follows that the images of a 3D point $\vec{X}$ by the two cameras are related as:
  - $\vec{x}' = P'\vec{X} = (K'R')(KR)^{-1}\underbrace{P\vec{X}}_{\vec{x}} = (K'R')(KR)^{-1}\vec{x}$, or $\vec{x}' = H\vec{x}$,
    - where $H = (K'R')(KR)^{-1}$
- **We have 2 cases to discuss about P and P'.**
  - Case 1: Just change the focal length (Zooming with no orientation)
    - Therefore $R = R'$
    - Thus $P' = K'K^{-1}P$
    - As such, $\vec{x}' = P'\vec{X} = K'K^{-1}P\vec{X} = K'K^{-1}\vec{x} \Rightarrow \vec{x}' = H\vec{x}$ $\quad (H_{3\times 3} = K'K^{-1}$ is our homography)
    - If $K = \begin{bmatrix} m_x f & 0 & x_0 \\ 0 & m_y f & y_0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$, $K' = \begin{bmatrix} m_x f' & 0 & x_0 \\ 0 & m_y f' & y_0 \\ 0 & 0 & 1 \end{bmatrix}$

      $K^{-1} = \begin{bmatrix} \frac{1}{\alpha_x} & 0 & \frac{-x_0}{\alpha_x} \\ 0 & \frac{1}{\alpha_y} & \frac{-y_0}{\alpha_y} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{m_x f} & 0 & \frac{-x_0}{m_x f} \\ 0 & \frac{1}{m_y f} & \frac{-y_0}{m_y f} \\ 0 & 0 & 1 \end{bmatrix}$

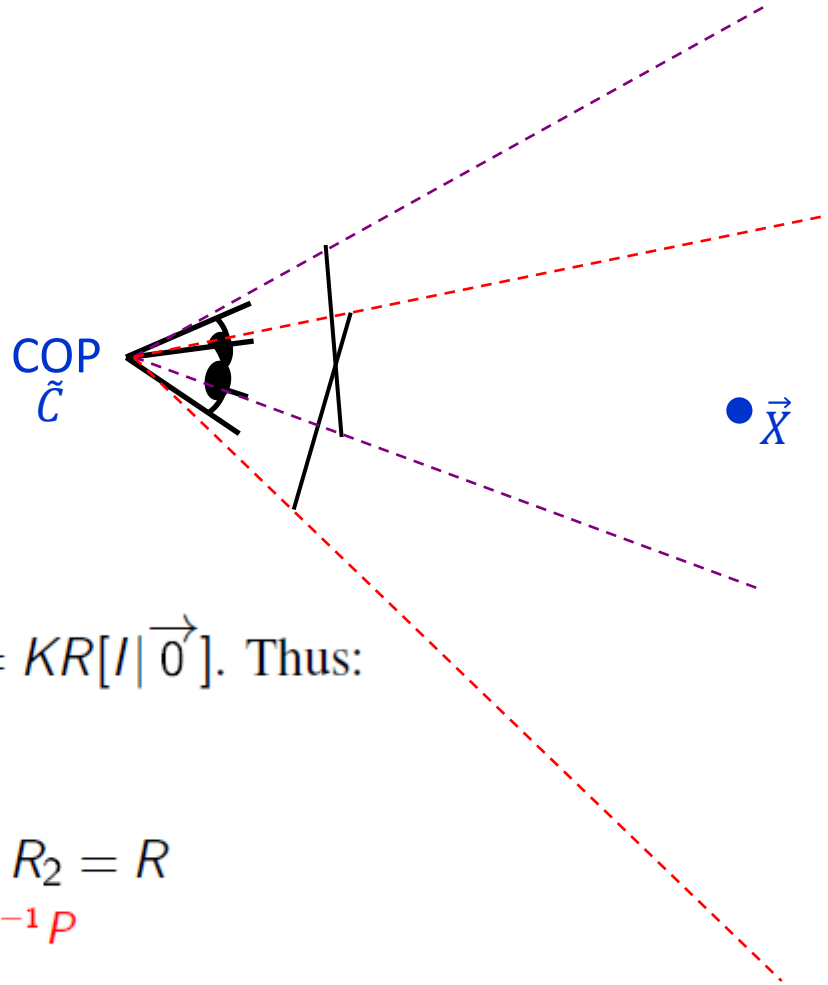
Mosaicking



- Case 1 (cont):
  - $\Rightarrow H = K'K^{-1} = \begin{bmatrix} m_x f' & 0 & x_0 \\ 0 & m_y f' & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{m_x f} & 0 & \frac{-x_0}{m_x f} \\ 0 & \frac{1}{m_y f} & \frac{-y_0}{m_y f} \\ 0 & 0 & 1 \end{bmatrix} =$

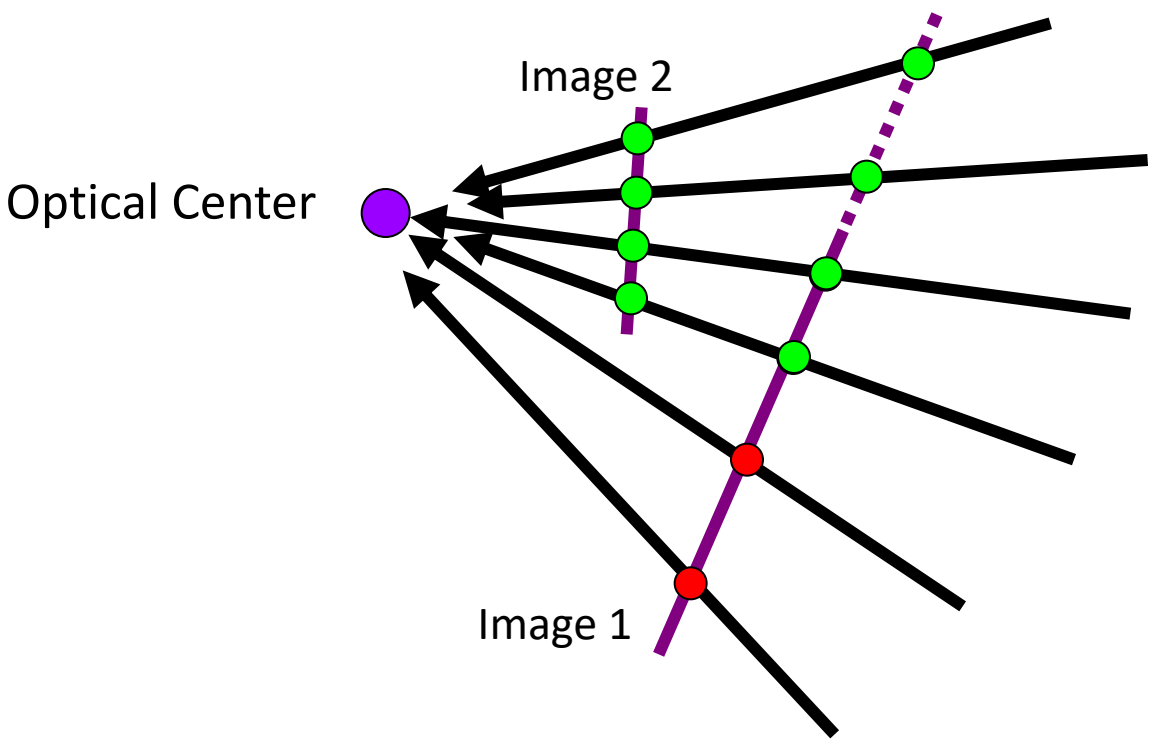
    $\begin{bmatrix} \kappa & 0 & (1-\kappa)x_0 \\ 0 & \kappa & (1-\kappa)y_0 \\ 0 & 0 & 1 \end{bmatrix}$, where $\kappa = \frac{f'}{f}$
    - So, the relation is a magnification factor through ratios of fpcal length

Mosaicking



- Case 2: Just rotate the camera (Pure Rotation)
  - First camera: $\vec{x} = P\vec{X} = KR_1\left[I|-\tilde{C}\right]\vec{X}$
  - Second camera: $\vec{x}' = P'\vec{X} = KR_2\left[I|-\tilde{C}\right]\vec{X}$
  - For the sake of our discussion let $\tilde{C} = \vec{0}$, $P = K[I|\vec{0}]$ and $P' = KR[I|\vec{0}]$. Thus:
    - First camera: $\vec{x} = P\vec{X} = K[I|0]\vec{X}$ (Since $\tilde{C} = \vec{0}$ and $R_1 = I$)
    - Second camera: $\vec{x}' = P'\vec{X} = KR[I|0]\vec{X}$ (Since $\tilde{C} = \vec{0}$ and $R_2 = R$)
  - Recall $P' = (K_2R_2)(K_1R_1)^{-1}P$, but $K_1 = K_2 = K$, $R_1 = I$ and $R_2 = R$
    - Therefore, $P' = (KR)(KI)^{-1}P \Rightarrow P' = (KR)I^{-1}K^{-1}P, \Rightarrow P' = KRK^{-1}P$
  - Thus: $\vec{x}' = P'\vec{X} = KRK^{-1}P\vec{X} = \underbrace{KRK^{-1}}_{H}\vec{x} = H\vec{x}$
    - $H = KRK^{-1}$ is the homography that relates the two images before and after the rotation.
    - $H = KRK^{-1}$ is called the conjugate rotation

What is the transformation?



Step 1: Convert pixels in image 2 to rays in camera 2's coordinate system.

Step 2: Convert rays in camera 2's coordinates to rays in camera 1's coordinates.

Step 3: Convert rays in camera 1's coordinates to pixels in image 1's coordinates.

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = \underbrace{KR^{-1}K}_{\text{3x3 homography}} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

How do we map points in image 2 into image 1?

	image 1	image 2
intrinsics	$\boldsymbol{K_1 = K}$	$\boldsymbol{K_2 = K}$
extrinsics (rotation only)	$\boldsymbol{R_1 = I_{3\times3}}$	$\boldsymbol{R_2 = R}$

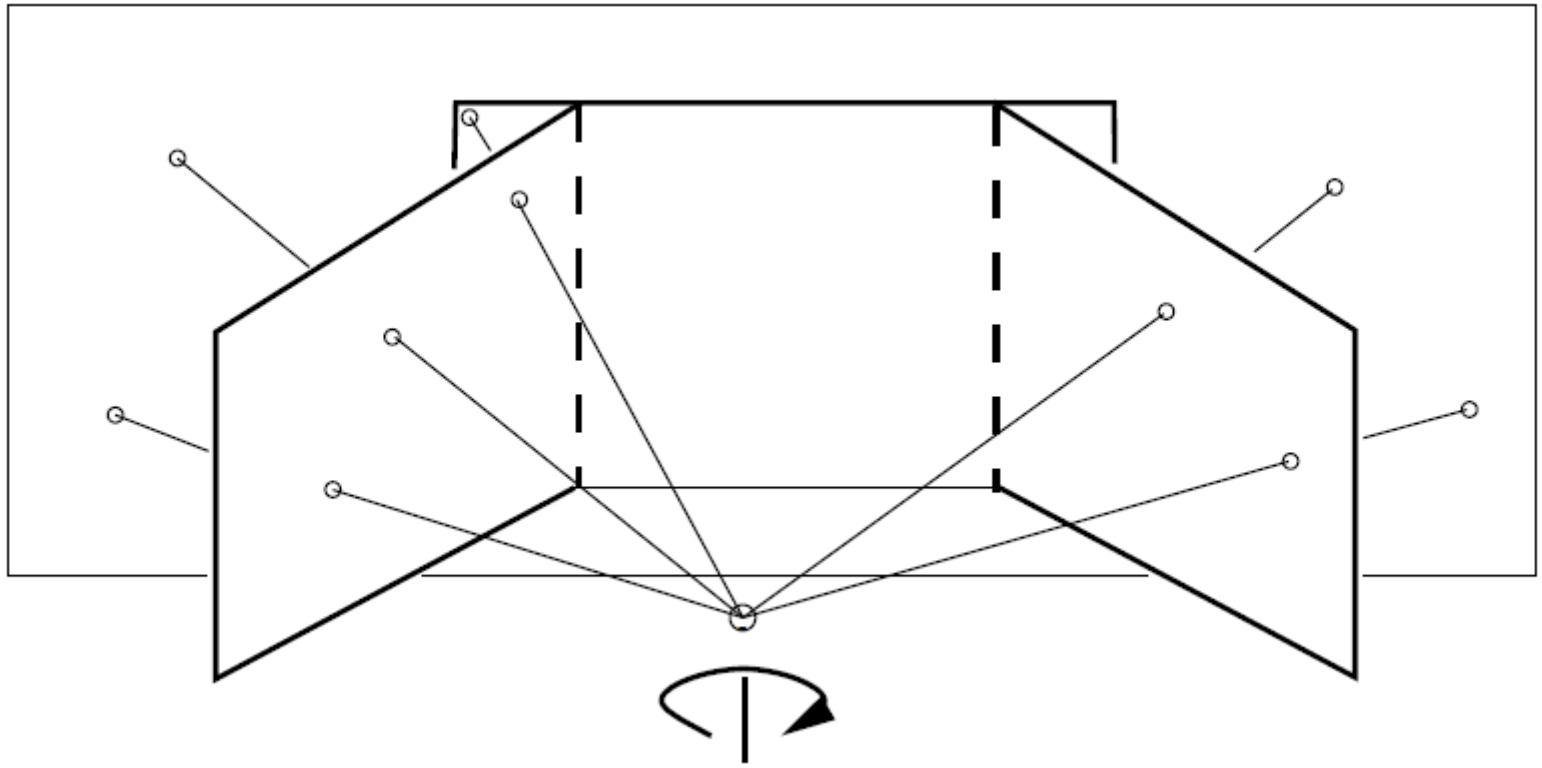
Mosaicking

- Case 2 (cont):
  - Note that R and KRK^{-1} are having the same eigenvalues but different eigenvectors.
  - Let $\vec{u}$ be the eigenvector of R
    - Thus, $R\vec{u} = \lambda\vec{u}$
  - Let $\vec{v}$ be the eigenvector of KRK^{-1}
  - Thus, $KRK^{-1}\vec{v} = \lambda\vec{v}$
    - If $R = \begin{bmatrix} cos\theta & 0 & sin\theta \\ 0 & 1 & 0 \\ -sin\theta & 0 & cos\theta \end{bmatrix}$, then the eigenvalues could be found from the characteristics polynomial of the matrix R given by $\begin{bmatrix} cos\theta - \lambda & 0 & sin\theta \\ 0 & 1-\lambda & 0 \\ -sin\theta & 0 & cos\theta - \lambda \end{bmatrix} = 0$
    - $\Rightarrow -(1-\lambda)\left[(cos\theta - \lambda)^2 + sin^2\theta\right] = 0$
    - Therefore, $\lambda_1 = 1$, $\lambda_2 = e^{i\theta}$, and $\lambda_3 = e^{-i\theta}$
    - It follows then: $\lambda_2 = cos\theta + i\,sin\theta$ and $\lambda_3 = cos\theta - i\,sin\theta$
    - Based on the above result: $\theta = cos^{-1}\left(\frac{\lambda_2 + \lambda_3}{2}\right)$

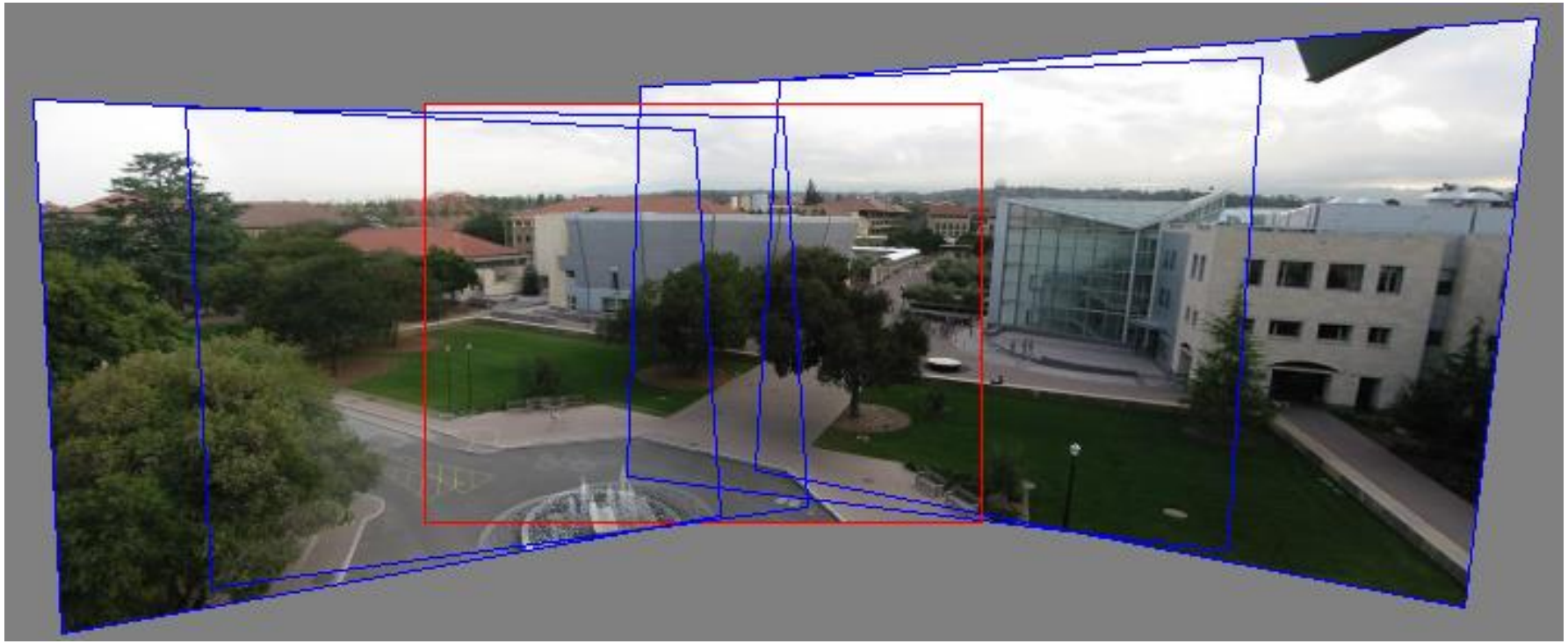
Mosaicking

- Case 2 (cont):
  - Therefore, to find the angle between two image views that are purely related through a rotation
    - You need to compute the homography $H = KRK^{-1}$ between the two images
    - Precondition the H matrix such that $det(H) = 1$
    - Apply the SVD to the conditioned homography H to find the eigenvalue decomposition i.e. the eigenvalues of the matrix R
    - From the eigenvalues estimate the angle θ between the two views.

Mosaicking



Mosaicking



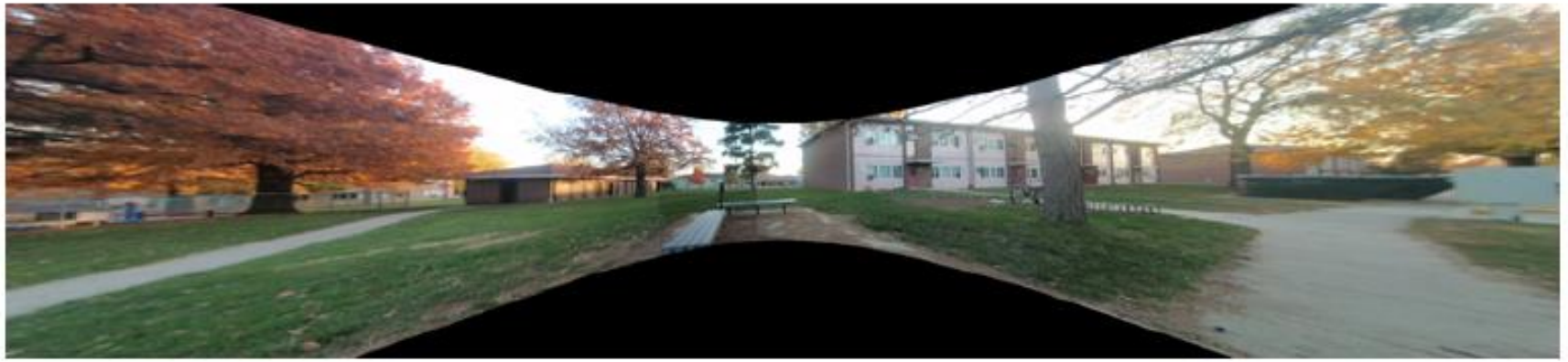
Example



	Theta1	Theta2	Theta3	Theta4	Theta5	Theta6	Theta7	Theta8	Theta9
Angle °	60.8214	42.7632	29.5776	16.7336	0	17.2893	30.7584	44.7454	64.6868

Example

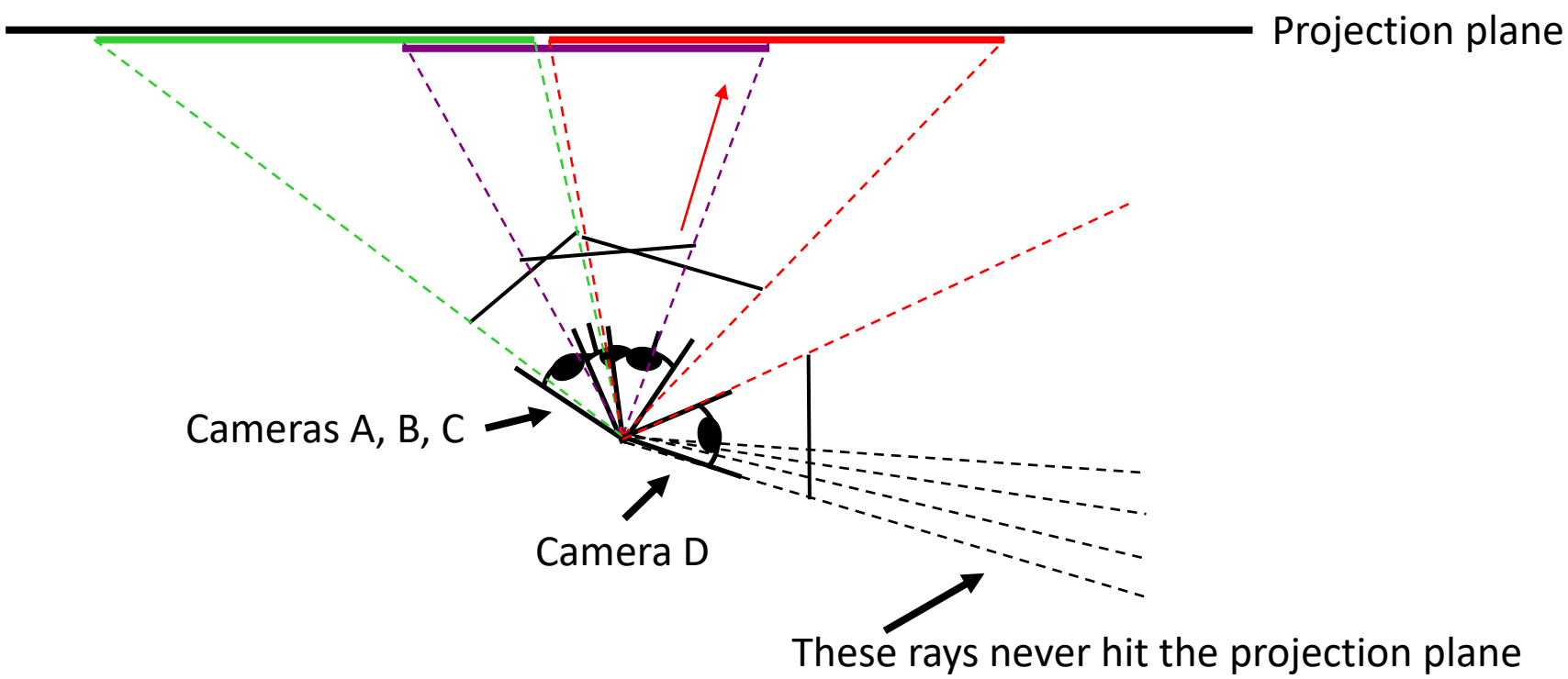




Mosaicking

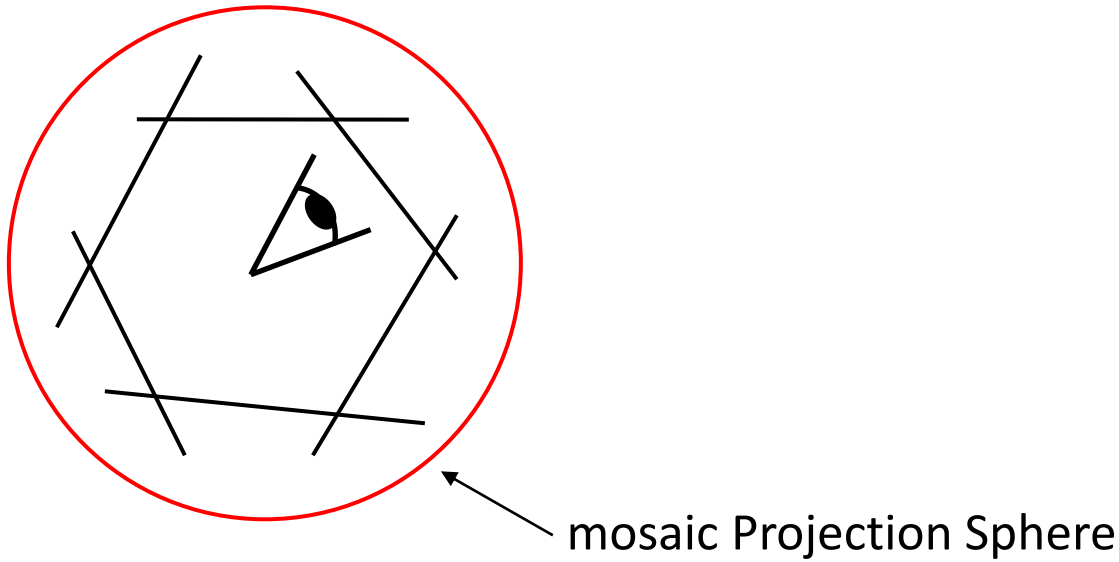
- **Now, coming back to our orginal question**
- **Can we use homography to create a 360 panorama?**

Answer: No


Projection plane
Cameras A, B, C
Camera D
These rays never hit the projection plane

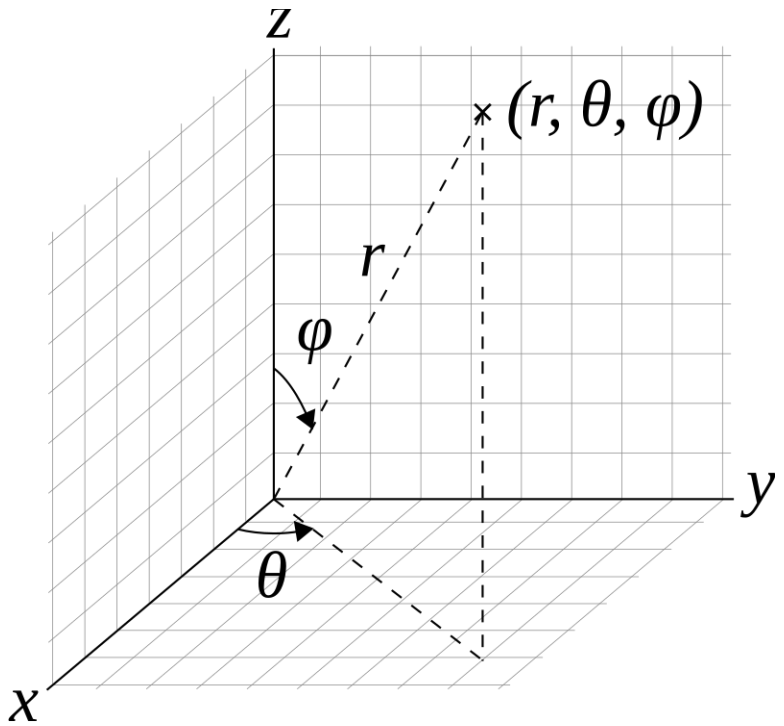
Panoramas

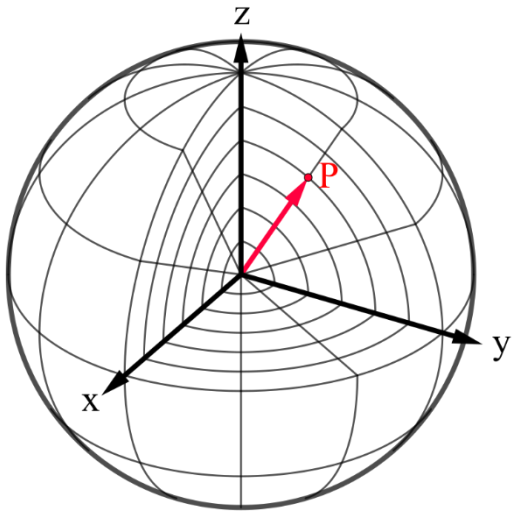
- What if you want a 360° field of view?



Necessary Background

- **Spherical Coordinates:** longitude/latitude
  - We represent points on a sphere using spherical coordinates (r, θ, ϕ), where:
    - θ: The azimuthal angle, ranging from 0 to 2π (0 to 360 degrees).
    - ϕ: The polar angle, ranging from 0 to π (0 to 180 degrees).

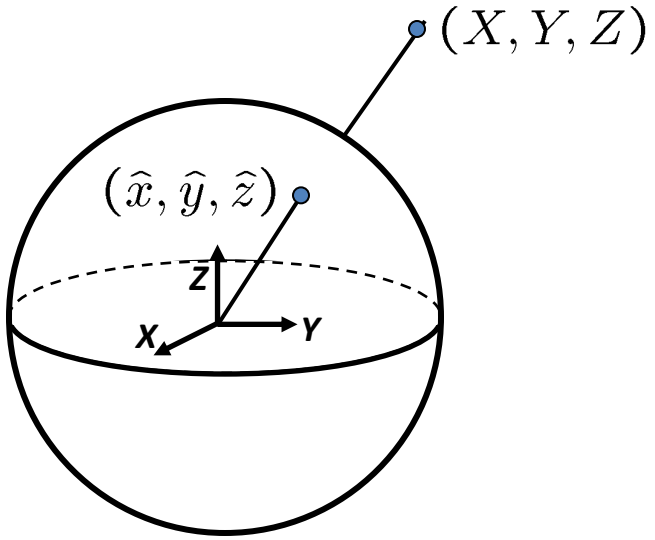




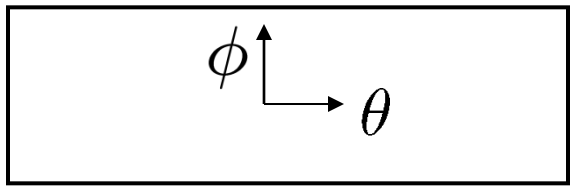
Source: https://en.wikipedia.org/wiki/Spherical_coordinate_system

Spherical projection:

You need to read more about this if you are interested



unit sphere



unwrapped sphere

- Map 3D point (X,Y,Z) onto a unit sphere (normalized)

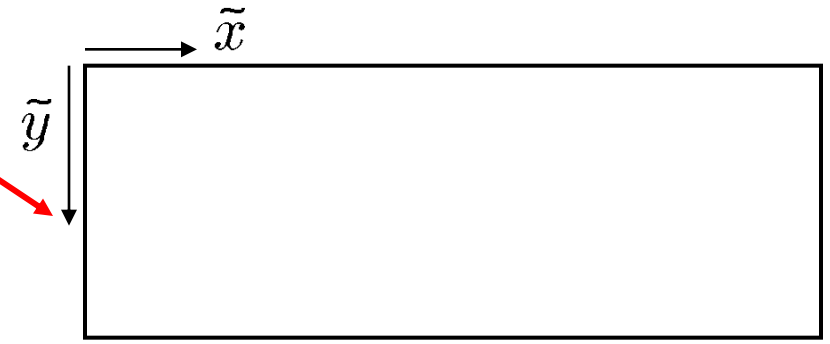
$$(\hat{x}, \hat{y}, \hat{z}) = \frac{1}{\sqrt{X^2+Y^2+Z^2}}(X, Y, Z)$$

- Convert to spherical coordinates: Calculate the azimuthal angle (θ) and polar angle (ϕ) using the following equations:

$$\theta = \tan^{-1}\frac{\hat{y}}{\hat{x}}, \phi = \cos^{-1}\hat{z}$$

- Convert to spherical image coordinates

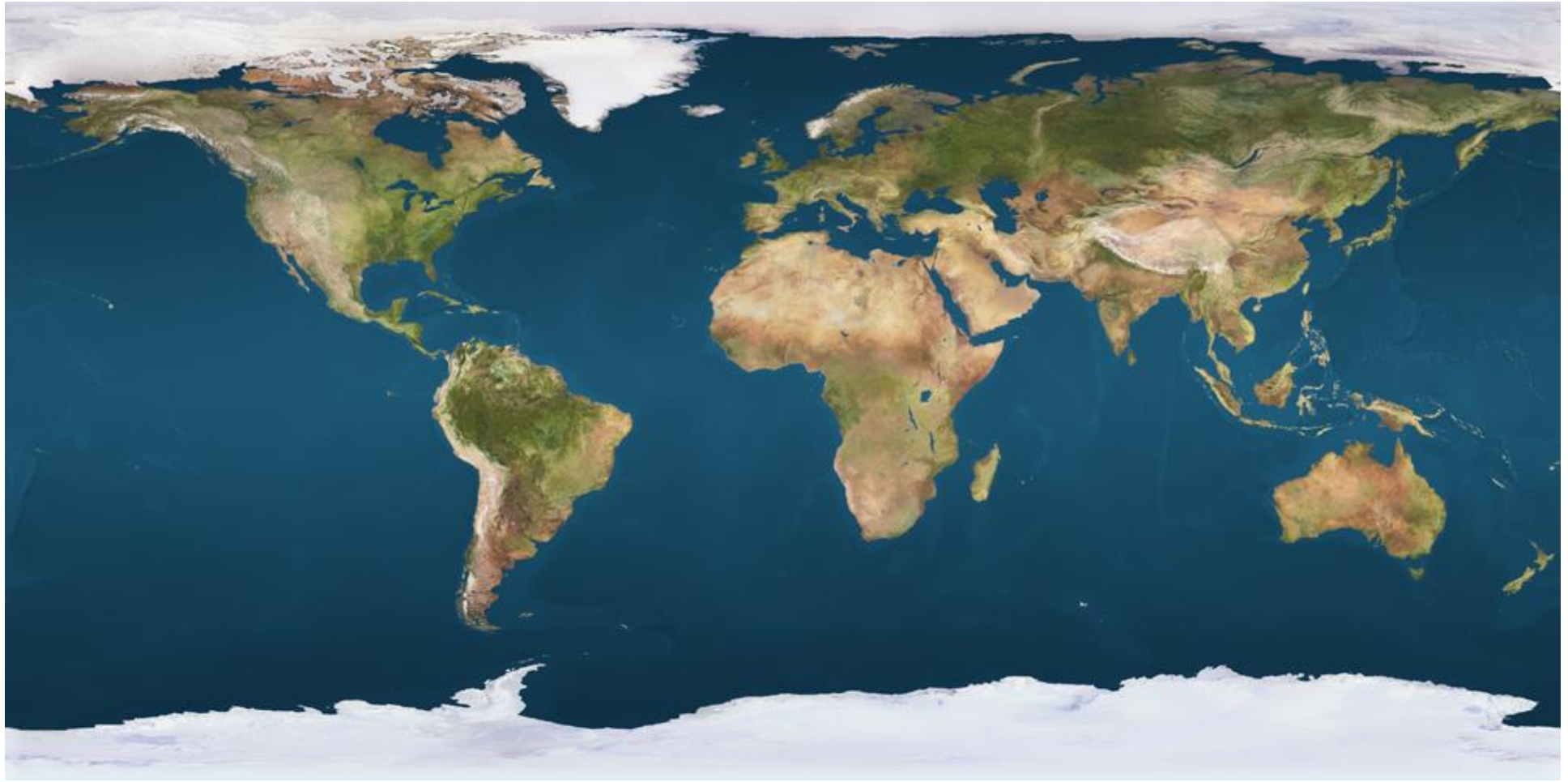
$\tilde{x}$= (θ / (2 * π)) * image_width
$\tilde{y}$= (φ / π) * image_height
where image_width and image_height are the dimensions of your spherical image.



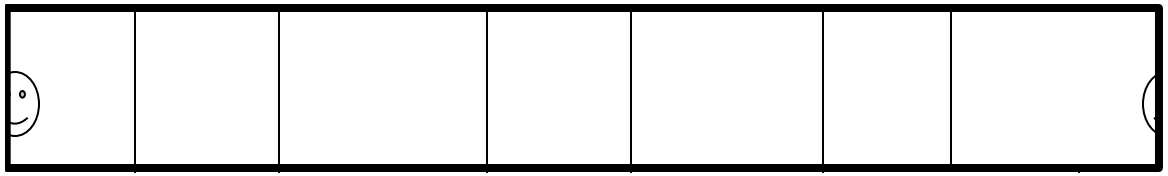
Spherical image

Unwrapping a sphere



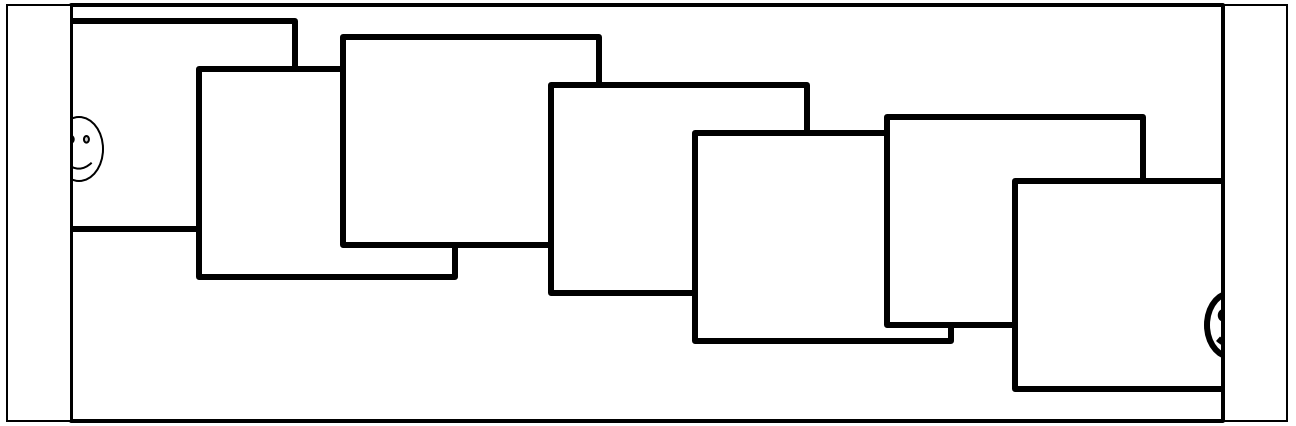


Assembling the panorama



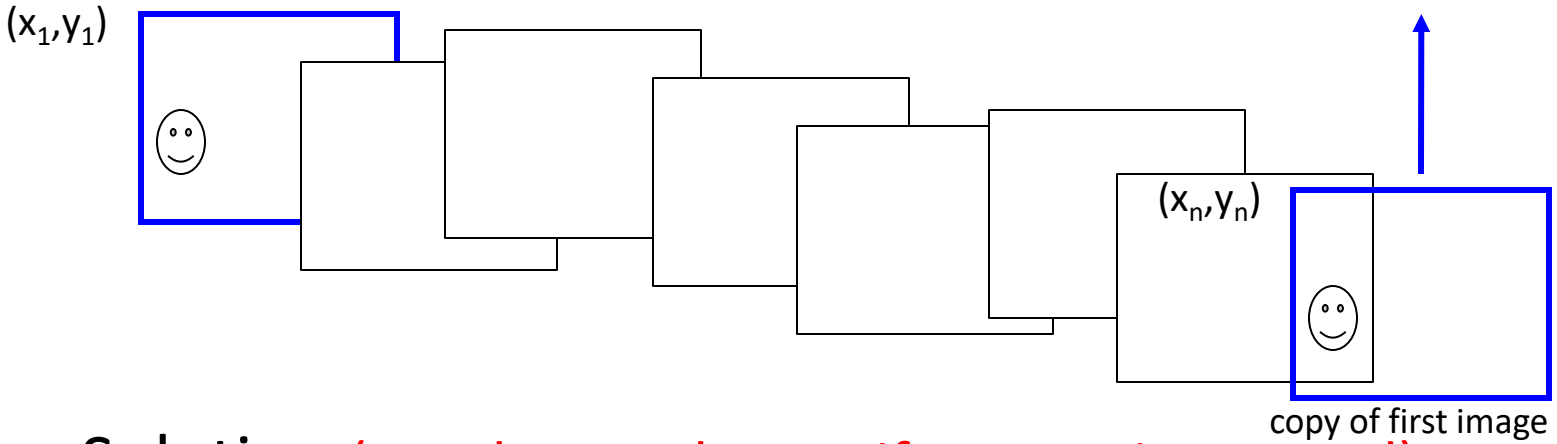
- Stitch pairs together, blend, then crop

Problem: Drift



- Error accumulation
  - small errors accumulate over time

Problem: Drift



- Solution (Need to read more if you are interested)
  - add another copy of first image at the end
  - this gives a constraint: $y_n = y_1$
  - there are a bunch of ways to solve this problem
    - add displacement of $(y_1 - y_n)/(n-1)$ to each image after the first
    - **apply an affine warp: y' = y + ax**
    - run a big optimization problem, incorporating this constraint
      - best solution, but more complicated
      - known as "bundle adjustment"

Blending

- We’ve aligned the images – now what?



Blending

- Want to seamlessly blend them together



- Two techniques:
  - Laplacian pyramid blending or feathering.
    - Read more about them if interested (implementation are found in OpenCV)

Some panorama examples

- Every image on Google Streetview





Questions?